

Research Discussion Assignment 1 by Sandra Dela Cruz

Part A

Now that we have covered basic techniques for recommender systems, choose one commercial recommender and describe how you think it works (content-based, collaborative filtering, etc). Does the technique deliver a good experience or are the recommendations off-target?

You may also choose one of the three non-personalized recommenders (below) we went over in class and describe the technique and which of the three you prefer to use.

1. Metacritic: How We Create the Metascore Magic
2. Rotten Tomatoes: About Rotten Tomatoes
3. IMDB: FAQ for IMDb Ratings

After reviewing the recommendation system used by Metacritic, Rotten Tomatoes, and IMDb, I believe Rotten Tomatoes had an advantage over the other two, even though IMDb and Metacritic are owned by the same company. Unlike personalized recommender systems that use content-based filtering or collaborative filtering, Rotten Tomatoes uses a non-personalized aggregation technique. It collects reviews from professional critics and audience members and summarizes them into two separate scores: the Tomatometer score (professional critics) and the Popcorn meter score (audience ratings). This approach makes it easier for users to compare expert opinions with general audience reactions. It allows users to decide for themselves which movies are worth watching and whether they align with their interests. Having both critic and audience scores provides a more balanced view and often helps make better decisions about what to watch.

Part B

Attacks on Recommender System

Read the article below and consider how to handle attacks on recommender systems. Can you think of a similar example where a collective effort to alter the workings of content recommendations have been successful? How would you design a system to prevent this kind of abuse?

Travis M. Andrews, The Washington Post (2017): Wisdom of the crowd? IMDb users gang up on Christian Bale's new movie before it even opens.

A similar example to the article Wisdom of the Crowd? IMDb Users Gang Up on Christian Bale's New Movie Before It Even Opens is the eMarketer article Instagram's Algorithm Clamps Down on Repurposed, Unoriginal Photos and Posts. As Instagram and other social media platforms have grown in popularity, many original content creators, artists, and small creators have struggled to gain audiences and followers because others steal their photos, videos, and ideas or simply repost their content without permission.

Meta, the parent company of Instagram, has taken steps to address this issue by monitoring accounts that repeatedly repost unoriginal content and limiting their reach. This helps ensure that original creators receive proper credit and have a better chance of being discovered by new audiences.

Similar to the IMDb example, groups of users can influence recommendation systems in ways that are unfair to creators. To prevent this type of abuse, I would design a system that identifies suspicious patterns, such as coordinated activity, excessive reposting, or large numbers of ratings from accounts with little activity history. The system could then reduce the influence of these actions and prioritize authentic user engagement. This would help create a fairer recommendation system and reduce manipulation.